

1. Administrative Overview & Directive

Ebrahim, this memorandum serves as the definitive technical blueprint and authoritative directive for the logical network design of the "Varsity Cyber Clash" E-Sports tournament. As the lead for technical execution, you are responsible for the precise implementation of the Variable Length Subnet Masking (VLSM) scheme detailed below.

This addressing strategy is built upon the 10.0.0.0/8 private network base and is strictly aligned with the "BAXZ1313 – Project Specification" and the final procurement lists identified in the project's costing documentation. You must adhere to these specifications without deviation to ensure network stability and to secure the maximum 10 marks allocated for the "Appropriate Private IP Addressing Scheme" in the project rubric.

2. Network Requirement Analysis

The "Varsity Cyber Clash" infrastructure must support a hybrid environment of stationary and wireless users. Based on the project directive and host density projections, the following host requirements have been prioritized to facilitate the VLSM process:

- **Spectators:** 300 hosts (High-density wireless access for event observers).
- **Gamers:** 70 hosts (Low-latency wired/wireless access for competitive play).
- **Admin:** 30 hosts (Broadcasters and event administrators requiring local resource access).
- **Server Farm:** 20 hosts (Critical infrastructure for game, web, and tournament services).
- **Management:** 5 hosts (Network infrastructure interfaces and core device management).

3. Master VLSM Addressing Plan

The following table defines the logical segmentation of the network. To demonstrate technical complexity (Rubric Item 6), note that while this design occupies the initial range of the 10.0.0.0/8 block, the remainder of the /8 address space is reserved for future campus-wide E-Sports expansion and institutional integration.

VLAN/Department	Network Address	Subnet Mask (CIDR & Decimal)	Usable IP Range	Default Gateway	Broadcast Address
Spectators	10.0.0.0	/23 (255.255.254.0)	10.0.0.2 – 10.0.1.254	10.0.0.1	10.0.1.255

Gamers	10.0.2.0	/25 (255.255.255.128)	10.0.2.2 – 10.0.2.126	10.0.2.1	10.0.2.127
Admin	10.0.2.128	/27 (255.255.255.224)	10.0.2.130 – 10.0.2.158	10.0.2.129	10.0.2.159
Server Farm	10.0.2.160	/27 (255.255.255.224)	10.0.2.162 – 10.0.2.190	10.0.2.161	10.0.2.191
Management	10.0.2.192	/29 (255.255.255.248)	10.0.2.194 – 10.0.2.198	10.0.2.193	10.0.2.199

Note: The first usable IP address in every subnet has been reserved specifically for the DIR-842 router interfaces to serve as the Default Gateway.

4. Hardware and Infrastructure Alignment

The logical plan must be mapped to the following physical assets identified in the Hardware Procurement List:

- **Primary Gateway:** One (1) **Router Wireless AC1200 (DIR-842)**. This device will manage the 500 Mbps Fiber Internet WAN link and perform inter-VLAN routing across all subnets.
- **Stationary Connectivity:** Two (2) **D-Link 24-Port Gigabit Switches (DGS-1024D)**. These will provide wired backhaul for the workstations and server farm.
- **Wireless Distribution:** Three (3) **TP-Link Access Points (TL-WA901ND)**. These must be strategically positioned to support the Spectator and Gamer wireless segments.
- **Endpoint Specifications:**
 - **Server Farm:** One (1) **Lenovo Think Station P720** (Intel Xeon E224G, 8GB ECC Memory) must be assigned a static IP (e.g., 10.0.2.162).
 - **Admin Subnet:** Twenty (20) **Dell Vostro 3681 Workstations** (Intel i5, 8GB RAM, 512GB SSD) must be addressed within the 10.0.2.130 – 10.0.2.158 range.